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GitHub Link for all code:

https://github.com/Kevin-Maniar/06Aug_Kevin_SE/tree/main/C%20_%20Assignment%20_%20Code

Module – 2 | Session – 1

Q – 1

Write a short paragraph in your own words defining what a 'program' is and give one real-life example from any app you use daily (like Instagram, Zomato, or Paytm).

Program: -

- ⇨ Program is a set of instructions written for a computer or mobile device to perform a specific task. Program tells a device what to do and how to do it.
- ⇨ For ex. Paytm – Allow user to make payments, receive payments, pay bills, and check for balance.

Q – 2

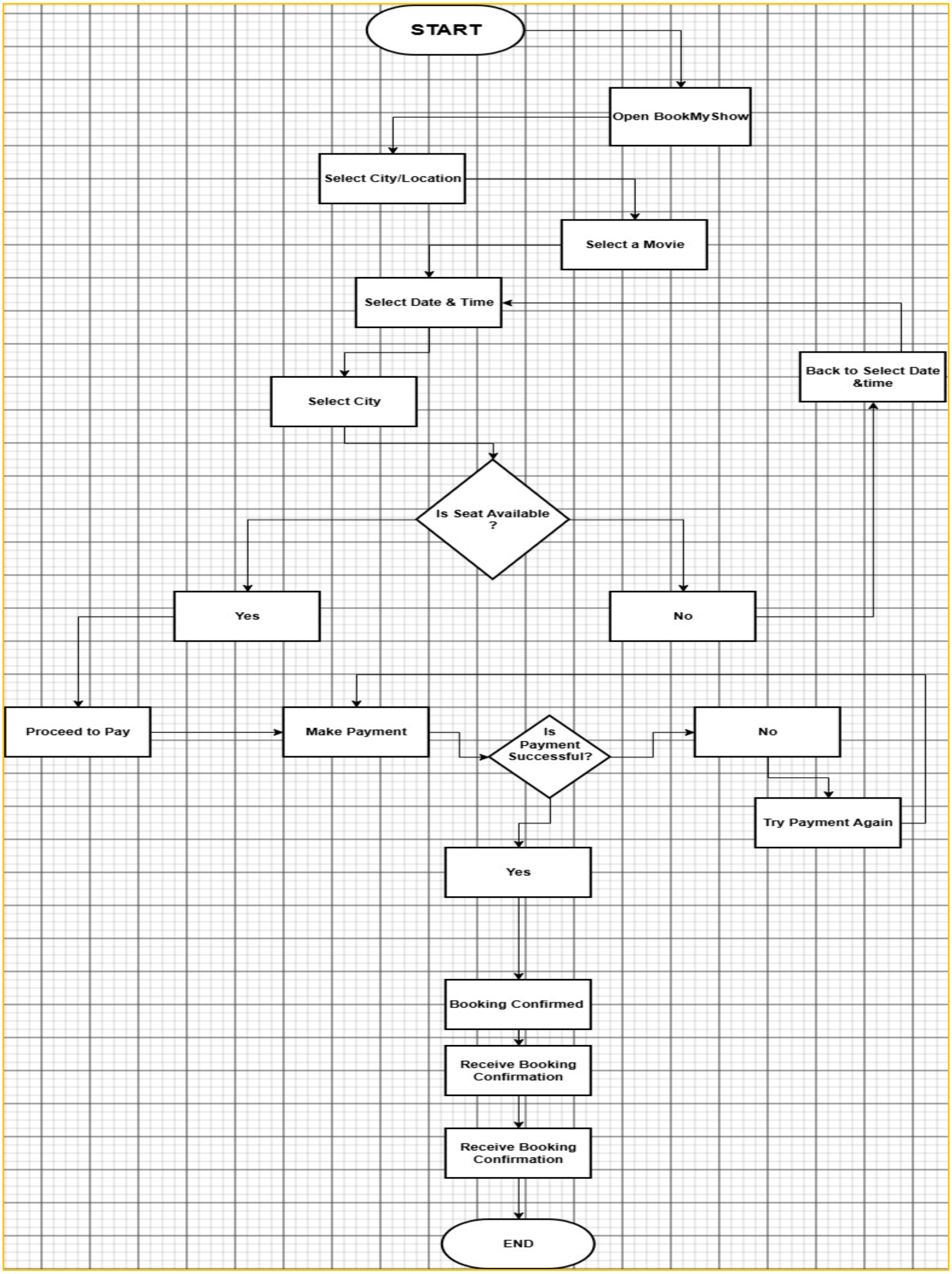
List three Programming languages you have heard of, and for each mention whether it is usually compiled, interpreted, or can be both.

Programming Languages: Java, Python, JavaScript, C++

- ⇨ Java: Compiled
- ⇨ Python: Interpreted
- ⇨ JavaScript: Both Compiled and Interpreted
- ⇨ C++: Compiled

Q – 3

Draw a flowchart (on paper or using any online tool) for the process of booking a movie ticket on BookMyShow, starting from opening the app to receiving the booking confirmation.


START
Open BookMyShow
Select City/Location
Select a Movie
Select Date & Time
Back to Select Date &time
Select City
Is Seat Available ?
Yes
No
Proceed to Pay
Make Payment
Is Payment Successful?
No
Try Payment Again
Yes
Booking Confirmed
Receive Booking Confirmation
Receive Booking Confirmation
END

Q – 4

Write down the steps (as an algorithm, using simple numbered instructions) for ordering food on Swiggy — from opening the app to completing payment.

- ⇨ Algorithm: Ordering food from swiggy

1. Start
2. Open the Swiggy App
3. Log in or Signup
4. Search for the Restaurant or select from the list
5. Select the food items you want to order
6. Add the selected items into the cart
7. Open the cart
8. Enter delivery location and confirm
9. Place order
10. Select the Payment method
11. Complete payment
12. If Payment is Successful, order is confirmed
13. Receive the order confirmation
14. End

Session – 2

Q – 1

Install GCC on your computer and verify the installation by running gcc --version in your terminal or command prompt.

Command Prompt

```
Microsoft Windows [Version 10.0.26200.8973]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Mr. Kevin>gcc --version
gcc (GCC) 16.2.0
Copyright (C) 2026 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.


C:\Users\Mr. Kevin>
```

Q – 2

Open VS Code or DevC++ and create a new file named hello.c. Write a C program that prints 'Hello, Instagram World!' to the console.

Hellooo.c

```c
#include<stdio.h>
int main()
{
    printf("Hello World!");
    return 0;
}
```

```
E:\06Aug_Kevin_SE\C _ Assign

Hello World!
--------------------------------
Process exited after 0.131 seconds with return value 0
Press any key to continue . . .
```

Q – 3

Identify and label the header, main function, and return statement in your hello.c program using comments.

```c
Hellooo.c

#include<stdio.h> // header file
int main() // main function where program start
{
    printf("Hello World!");
    return 0; // return statement
}
```

Q – 4

Compile your hello.c file using the GCC command line (gcc hello.c -o hello) and run the output file to display the message in your terminal.

```
Command Prompt

C:\Users\Mr. Kevin>cd /d E:\06Aug_Kevin_SE\C _ Assignment _ Code

E:\06Aug_Kevin_SE\C _ Assignment _ Code>Hellooo
Hello World!
I am gcc Compiler
E:\06Aug_Kevin_SE\C _ Assignment _ Code>
```

Session: 3

Q – 1 Declare variables for a Flipkart product: productName (as a string), price (float), and rating (double). Assign sample values and print each variable with its data type.

S3T1_FP.c

```c
#include <stdio.h>

#define get_type(X) _Generic((X), \
    float: "float", \
    double: "double", \
    char*: "string ", \
    default: "unknown type")

int main()
{
    char productName[] = "Samsung Galaxy S24";
    float price = 59999.99;
    double rating = 4.5;

    printf("Product Name: %s\n", productName);
    printf("Data Type: %s\n\n", get_type(productName));

    printf("Price: %.2f\n", price);
    printf("Data Type: %s\n\n", get_type(price));

    printf("Rating: %.1lf\n", rating);
    printf("Data Type: %s\n", get_type(rating));

    return 0;
}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Product Name: Samsung Galaxy S24
Data Type: string

Price: 59999.99
Data Type: float

Rating: 4.5
Data Type: double

--------------------------------
Process exited after 0.06464 seconds with return value 0
Press any key to continue . . .
```

Q – 2

Create a constant variable to store the GST rate (for example, 18%) and use it to calculate the final price of a Zomato order with a given base price.
Constraint: The GST rate must not be changeable after its initial assignment.

S3T1_FP.c S3T2_Const.cpp

```c
#include<stdio.h>
int main ()
{
    const float GST_RATE = 0.18; //Gst rate 18%
    float base_price ;

    printf("Enter Base price of food:");
    scanf("%f",&base_price);

    double gst = GST_RATE * base_price ;
    float final_price = base_price + gst;

    printf("final price is :%.2f",final_price);
    return 0;
}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Enter Base price of food:1200
final price is :1416.00
--------------------------------
Process exited after 5.663 seconds with return value 0
Press any key to continue . . .
```

Q – 3

Write a program that stores your favorite Spotify playlist's name (string), total number of songs (int), and average song duration in minutes (float). Print all values in a single formatted sentence.

```c
/*Write a program that
stores your favorite Spotify playlist's name (string),
total number of songs (int),
and average song duration in minutes (float).
Print all values in a single formatted sentence.*/

#include<stdio.h>
int main()
{
    char playlist[10];
    int total_songs;
    float avg_song_duration;

    printf("Enter Your fav Spotify playlist's Name:");
    scanf("%s",&playlist);
    printf("total_songs are :");
    scanf("%d",&total_songs);
    printf("Avg Duration:");
    scanf("%f",&avg_song_duration);

    printf("Playlist Name: %s\n",playlist);
    printf("Total Songs : %d\n",total_songs);
    printf("Total Duration: %.0f Minutes\n",avg_song_duration);

    return 0;

}
```

```
Enter Your fav Spotify playlist's Name:Chill_Vibes
total_songs are :20
Avg Duration:4
Playlist Name: Chill_Vibes
Total Songs : 20
Total Duration: 4 Minutes

--------------------------------
Process exited after 17.47 seconds with return value 0
Press any key to continue . . .
```

Q – 4

Identify and correct the invalid variable names from this list:
1stPlayer, player_age, $score, total-marks, userName.

- ⇨ Invalid Variable names are : 1stPlayer , $score , total-marks
- ⇨ Correct Version : first_player , score , total_marks

Session: 4

Que – 1

Create a simple JavaScript function called calculateTotal that takes two numbers: itemPrice and quantity and returns the total bill amount using arithmetic operators.

S3T3_Playlist.cpp S4T1_Arithmatic_operator.cpp

```c
/*Create a simple JavaScript function called
calculateTotal that takes two numbers:
itemPrice and quantity, and returns the
total bill amount using arithmetic operators.*/
#include<stdio.h>
int main()
{


    float itemPrice;
    int Quantity;
    float Cal_Total;

    printf("Enter Item Price:");
    scanf("%f",&itemPrice);
    printf("Enter Quantity:");
    scanf("%d",&Quantity);

    Cal_Total = itemPrice * Quantity;
    printf("Total Bill Amount is: %.2f INR",Cal_Total);

    return 0;
}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Enter Item Price:120
Enter Quantity:3
Total Bill Amount is: 360.00 INR
--------------------------------
Process exited after 2.473 seconds with return value 0
Press any key to continue . . .
```

Q – 2 Build a Flipkart-style discount calculator given product price, discount percentage, and a boolean isMember, use arithmetic and logical operators to calculate the final price (apply an extra 5% off if isMember is true).

S4T2_Discount_Calculator.cpp

```c
#include<stdio.h>
int main()
{
    /*Build a Flipkart-style discount calculator:
    given product price,
    discount percentage,
    and a boolean isMember,
    use arithmetic and logical
    operators to calculate the final price
    (apply an extra 5% off if isMember is true). */

    float product_price, discount_percentage;
    float final_price, discount_price;
    int isMember;

    printf("Enter Product Price:");
    scanf("%f",&product_price);

    printf("Enter Discount Percentage:");
    scanf("%f",&discount_percentage);

    printf("Are You Member: (1 for Yes , 0 for NO)");
    scanf("%d",&isMember);

    // Count Normal price
     discount_price = product_price * (discount_percentage/100);
    //Price After Discount
    final_price = product_price - discount_price;
    //Extra 5% discount

    if (isMember == 1)
    {
        final_price = final_price -(final_price *5/100);
        printf("Your Final price is :%.2f",final_price);
    }
    else
    {
        printf("Final Price = %.2f$",final_price);
    }

}
```

```
E:\06Aug_Kevin_SE\C _ Assign

Enter Product Price:50000
Enter Discount Percentage:20
Are You Member: (1 for Yes , 0 for NO)1
Your Final price is :38000.00
--------------------------------

Process exited after 10.04 seconds with return value 0
Press any key to continue . . .
```

Q – 3

Write a function isEligibleForOffer that takes a user's age and total order value and returns true if the user is 18 or older AND the order value is above 500, otherwise false.

Hint: Use relational and logical operators together.

S4T2_Discount_Calculator.cpp S4T3_Relational_logical_oper.cpp

```c
    /*Write a function isEligibleForOffer
    that takes a user's age and total order value,
    and returns true if the user is 18 or older AND
    the order value is above 500,
    otherwise false.*/

#include<stdio.h>

bool isEligibleForOffer (int age,float order_value)
{
    if(age>=18  && order_value>500)
    {
        return true;
    }
    else{
        return false;
    }
}
int main()
{
    int age;float order_value;
    printf("Enter Your Age:-");
    scanf("%d",&age);

    printf("Enter Your Total Order Value:-");
    scanf("%f",&order_value);

    if(isEligibleForOffer(age,order_value))
    {
        printf("YOU ARE ELIGIBLE FOR OFFER");
    }
    else{
        printf("YOU ARE NOT ELIGIBLE FOR OFFER!");
    }
    return 0;
}
```

```
E:\06Aug_Kevin_SE\C _ Assign

Enter Your Age:-18
Enter Your Total Order Value:-5000
YOU ARE ELIGIBLE FOR OFFER
--------------------------------
Process exited after 5.618 seconds with return value 0
Press any key to continue . . .
```

Q – 4

Given three variables: likes, comments, and shares (all numbers), write code to check if a post is 'trending' on Instagram (at least 1000 likes OR more than 200 comments AND at least 50 shares). Print the result.

```
S4T4_.cpp

    /* Given three variables: likes, comments, and shares (all numbers),
    write code to check if a post is 'trending' on
    Instagram (at least 1000 likes OR more than
    200 comments AND at least 50 shares).
    Print the result. */

#include<stdio.h>
main ()
{
    int likes,comments,share;
    printf("Enter Total Number of Likes:");
    scanf("%d",&likes);
    printf("Enter Total Number of Comments:");
    scanf("%d",&comments);
    printf("Enter Total Number Share:");
    scanf("%d",&share);
    printf("-----------\n");
    printf("Liked By %d Fan \ncommented %d Fan \nShare By %d Fan\n",likes,comments,share);

    if(likes>=1000 || comments>200 && share>=50)
    {
        printf("-----------\n");
        printf("Result: Your Post is Trending! Congratulations");
    }
    else
    {
        printf("-----------\n");
        printf("Result: Your post is not Trending!!!");
    }
}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Enter Total Number of Likes:300
Enter Total Number of Comments:400
Enter Total Number Share:51
-----------
Liked By 300 Fan
commented 400 Fan
Share By 51 Fan
-----------
Result: Your Post is Trending! Congratulations
--------------------------------
Process exited after 29.73 seconds with return value 0
Press any key to continue . . .
```

Q – 5

Write a code snippet that demonstrates the difference between pre-increment (++count) and post-increment (count++) by logging the values before and after using both on a followerCount variable.

S4T4_.cpp S4T5_.cpp

```c
    /*  Write a code snippet that
    demonstrates the difference between pre-increment
    (++count) and post-increment (count++)
    by logging the values before and after using both on a
    followerCount variable. */
#include<stdio.h>
int main()
{
    int follower = 100;

    printf("Pre Increment Follower ---> %d\n",++follower);
    printf("Pre Increment Follower ---> %d\n",++follower);
    printf("Pre Increment Follower ---> %d\n",++follower);

    printf("Post Increment Follower ---> %d\n",follower++);
    printf("Post Increment Follower ---> %d\n",follower++);
    printf("Post Increment Follower ---> %d\n",follower++);

    /* follower++ takes 100 first and then add +1
    ++follower add +1 in exist follower number

    pre increment give result by +1 and
    post increment give result by using old value first and then +1*/
    return 0;
}
```

E:\06Aug_Kevin_SE\C _ Assign

```
Post Increment Follower ---> 100
Post Increment Follower ---> 101
Post Increment Follower ---> 102
Pre Increment Follower ---> 104
Pre Increment Follower ---> 105
Pre Increment Follower ---> 106

--------------------------------
Process exited after 0.444 seconds with return value 0
Press any key to continue . . .
```